

SQL IN MANUAL TESTING WORKFLOW

Signup validation, authentication verification, profile state updates, and user inactivity auditing.

DUPLICATE CHECK
Pre-Signup

POST-SIGNUP
Row Insertion

AUTH LOGIC
Hash & Status

STATE AUDIT
Inactivity Query

1. Duplicate Email Check Before Signup

PRE-SIGNUP VERIFICATION

Query the users table to ensure duplicate accounts are prevented before creating a new user record.

```
SELECT EXISTS (  
  SELECT 1  
  FROM users  
  WHERE email = 'aarav.sharma@example.com'  
) AS email_exists;
```

Expected Verification Output:

email_exists	System Action	User Experience
1 (True)	Reject registration request	Display "Email already registered" on UI
0 (False)	Proceed with account creation	Create new user record in DB

2. Post-Signup Data Persistence Verification

DATA INTEGRITY

Verify that registered user details (username, email, phone) match backend records immediately after form submission.

```
SELECT user_id, username, email, phone, created_at  
FROM users  
WHERE username = 'aarav_s'  
  AND email = 'aarav.sharma@example.com'  
  AND phone = '9876543210';
```

Verification Result Set:

user_id	username	email	phone	created_at
108	aarav_s	aarav.sharma@example.com	9876543210	2026-09-23 10:30:00

3. Authentication Logic & Account Status Testing

AUTH BACKEND

Fetch stored password hashes and account status to verify backend authentication rules during login testing.

```
SELECT user_id, username, password_hash, account_status
FROM users
WHERE username = 'aarav_s';
```

Result Verification:

user_id	username	password_hash	account_status
108	aarav_s	\$2b\$12\$elmiTXuWVxfM37uY4JANjO...	ACTIVE

4. Profile Update Simulation & Confirmation

UI UPDATE TEST

Simulate an Instagram-style profile picture and bio update, followed by a verification SELECT query.

```
UPDATE users
SET profile_pic_url = 'https://cdn.app.com/profiles/u108_v2.jpg',
    bio = 'Software Engineer | Tech Enthusiast 🚀',
    updated_at = CURRENT_TIMESTAMP
WHERE user_id = 108;

SELECT user_id, profile_pic_url, bio, updated_at
FROM users WHERE user_id = 108;
```

Verification Result Set:

user_id	profile_pic_url	bio	updated_at
108	https://cdn.app.com/profiles/u108_v2.jpg	Software Engineer Tech Enthusiast 🚀	2026-09-23 12:20:00

5. Inactive Users Post-Release Database Audit

POST-RELEASE AUDIT

Audit the newly added last_login timestamp column to identify accounts inactive for over 30 days or never logged in.

```
SELECT user_id, username, email, last_login
FROM users
WHERE last_login < NOW() - INTERVAL '30 days'
    OR last_login IS NULL
ORDER BY last_login ASC;
```